

Benign recurrent intrahepatic cholestasis type 2

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Pipeline complete — awaiting your sign-off

Repurposing hypothesis: VERAPAMIL HYDROCHLORIDE → Benign recurrent intrahepatic cholestasis type 2

Repurposing-only pool: the candidate compounds for this target were restricted to FDA-approved / known drugs (ChEMBL max_phase ≥ 4) at collection time. Unapproved research-grade tool compounds were excluded from the pool, not merely down-ranked.

1. Hypothesis summary

VERAPAMIL HYDROCHLORIDE is proposed as a repurposing candidate against **Benign recurrent intrahepatic cholestasis type 2** via the target **ABCB11**. It shows a ChEMBL median pChEMBL affinity of 4.69 at assay confidence 9/9, an Open Targets target-disease association of 0.858, and a Tanimoto similarity of 0.979 to VERAPAMIL. Target network context (BioGRID, physical/genetic — not mechanism): NR4A1, LY6D, AP2A1, Ap2a1, Ap2m1, DDIT4L, HIST2H3PS2, ORF3a. The resulting composite score is 0.7409.

Chemist rationale: Verapamil hydrochloride has a median pChEMBL affinity of 4.69 against ABCB11, measured at assay confidence score 9, and has a Tanimoto similarity of 0.979 to the approved drug

verapamil. The target ABCB11 has the following BioGRID physical/genetic interactors on record: NR4A1, LY6D, AP2A1, Ap2a1, Ap2m1, DDIT4L, HIST2H3PS2, and ORF3a.

STAGE 1 PRIORITIZATION SCORES

- **tractability_score:** 0.5908 (ChEMBL bioactivity + AlphaFold pLDDT – prior-trial-failure penalty)
- **unmet_need_score:** 1.0000 (treatment availability + prevalence)

These are computed by the same formulas used to rank the full rare-disease / NTD universe; a manually chosen target is scored identically, never faked or skipped.

2. Evidence table

Evidence	Value
ChEMBL median pChEMBL affinity	4.69
Assay confidence score (0-9)	9
Open Targets association score	0.858
Tanimoto to nearest approved drug	0.979 (VERAPAMIL)
Approved / known drug	yes
Mutation-specific approved indication (disclosure)	No specific mutation named in approved indication
Lipinski/Veber (MW, logP, HBD, HBA, TPSA, rotB)	491.1, 5.51, 0, 6, 64.0, 13
Lipinski violations / Veber pass	1 / no
AFDB apo structure mean pLDDT (free protein, no ligand)	83.1
Boltz structure confidence (0-1)	0.749
Boltz binding-pose confidence (0-1)	0.558
Boltz predicted affinity (relative optimization score, 0-1, NOT a Kd)	0.186

Evidence	Value
Boltz predicted structure (CIF)	Download CIF (/api/structures/0837a877d78b4b2862429bfc13042df8.cif)
Boltz ADME – lipophilicity (logD)	4.647
Boltz ADME – permeability	0.512
Boltz ADME – solubility	high-risk
openFDA adverse-event signal (FAERS)	none reported
Prior trials for this exact drug+disease	0
Target discovery method	genetic_association

3. Full source citations

- **PMIDs (7):** 15300568, 18853996, 26223708, 29507376, 30899697, 32647738, 34013234
- **ChEMBL activity IDs (1):** 18081202
- **NCT numbers (0):** none

4. Composite score breakdown

Term	Weight	Component value	Contribution
Normalized pChEMBL affinity	0.30	0.241	0.0724
Assay confidence (score / 9)	0.20	1.000	0.2000
Normalized Open Targets association	0.20	0.858	0.1716
Normalized Tanimoto similarity	0.15	0.979	0.1469
No prior failed trial (1/0)	0.15	1	0.1500
Composite (weighted sum - penalty - cap)			0.7409

Weighted sum before penalty = 0.7409; penalty = 0.0000; reported composite_score = 0.7409.

5. Limitations

- **Binding is not efficacy.** A high binding-pose confidence (0.558) or predicted affinity (0.186) only suggests the molecule may occupy the target; it does NOT establish agonism vs. antagonism, functional modulation, or therapeutic benefit.
- **ADME values are model predictions, not measurements.** The Boltz lipophilicity/permeability/solubility numbers are computed estimates and must be confirmed experimentally before any decision.
- **Structure confidence is bounded.** This hypothesis relies on a Boltz structure_confidence of 0.749 (complex pLDDT 0.744) and an AFDB apo mean pLDDT of 83.1; the AFDB model contains NO ligand, so the protein-ligand pose is entirely a Boltz prediction.
- **Assay-type and species caveats.** The affinity is a median pChEMBL over Homo sapiens IC50/Ki assays at confidence ≥ 8 ; assay heterogeneity and the bounded approved-drug reference set for Tanimoto still apply.
- **Absence of evidence is not evidence of absence.** A zero prior-trial count or no adverse-event signal may reflect that the pair has simply never been studied, not that it is safe or untried.
- **This is a repurposing *hypothesis*, not a finding.** It is a prioritised starting point that requires wet-lab and, ultimately, clinical validation.

TARGET DRUGGABILITY CONTEXT

- **No approved drug currently exists with a known mechanism against this target** (ChEMBL mechanism endpoint, Homo sapiens only).
- **Historical difficulty literature:** insufficient signal found (fewer than 2 qualifying abstracts in targeted PubMed searches for undruggability / resistance / difficulty).

Druggability context is informational only. It does not affect tractability_score, unmet_need_score, composite_score, or STRONG_MATCH.

HUMAN REVIEW CHECKPOINT

This is a falsifiable hypothesis, not a finding.

Record your scientific reasoning below, then sign off. Approving advances this candidate for wet-lab validation; rejecting closes the case permanently.

REVIEWER NOTE **required**

e.g. Affinity and structure confidence justify wet-lab follow-up despite the sub-threshold composite score.



Approve case

Reject case